

Project : Loan Default Risk

Note target = 1 means not approved and target = 0 means loan approved

1. **Retrieve the total number of loan applications received**
`select count(*) as total_application from loan_data;`
2. Find the number of applications for each loan status (Approved, Refused, Cancelled, Unused Offer).
`select
case
when Target=0 then 'Approved'
when Target=1 then 'Refused'
else 'unknown'
end as Loan_status,
count(*) as Num_application from loan_data
group by case
when Target=0 then 'Approved'
when Target=1 then 'Refused'
else 'unknown'
end;`
3. Display the number of unique clients who have applied for a loan
`select count(distinct SK_ID_CURR) as Unique_clients from loan_data;`
4. Find the average loan amount requested by applicants.
`select round(avg(AMT_CREDIT),2) as Avg_Loan_Amt from loan_data;`
5. List the top 10 applicants who applied for the highest loan amounts.
`select * from loan_data order by AMT_CREDIT desc limit 10;`
6. Find the approval rate of loans (Approved / Total Applications).
`select round(
(sum(case when target=0 then 1 else 0 end)*100)/count(*),2) as Approval_Rate from
loan_data;`
7. Determine the percentage of loans that were refused or cancelled.
`select round(sum(case when target=1 then 1 else 0 end)*100 / count(target),2) as
percentage_of_refuse from loan_data;`

8. Identify which loan type (e.g., consumer loan, cash loan, etc.) has the highest approval rate.
 select NAME_CONTRACT_TYPE ,
 round((sum(case when target=0 then 1 else 0 end)*100)/count(*),2) as Approval_Rate from
 loan_data group by
 NAME_CONTRACT_TYPE order by NAME_CONTRACT_TYPE desc;
9. Find the average loan amount for approved vs. refused loans.
 select target , round(avg(AMT_CREDIT)) as Avg_Loan from loan_data group by target;
10. Display the month-wise trend of loan applications and approvals.
 select WEEKDAY_APPR_PROCESS_START as Week_Day , count(*) as Loan_application ,
 sum(case when target=0 then 1 else 0 end) as Approvals_Application from loan_Data
 group by WEEKDAY_APPR_PROCESS_START order by
 field(WEEKDAY_APPR_PROCESS_START ,
 'Monday','Tuesday','Wednesday','Thursday','Friday','Saturday','Sunday');
11. Find the distribution of applicants by income range (e.g., <50K, 50K–100K, >100K).
 select
 case when AMT_INCOME_TOTAL < 50000 then '50K'
 when AMT_INCOME_TOTAL between 50000 and 100000 then '50K-100K'
 else '>100K'
 end as Income_Range,
 count(*) as Total_application from loan_data
 group by Income_Range
 order by
 case
 when Income_Range = '50K' then 1
 when Income_Range = '50K-100K' then 2
 when Income_Range = '>100K' then 3
 End ;
12. Calculate the average loan amount and income for applicants who defaulted vs. those who didn't.
 select TARGET , round(avg(AMT_INCOME_TOTAL),2) as Avg_Income ,
 round(avg(AMT_CREDIT),2) as avg_loan from loan_data group by target;
13. Identify which occupation type or employment status has the highest default rate.
 select OCCUPATION_TYPE ,
 round((sum(case when target=1 then 1 else 0 end)*100)/count(*),2) as Default_Rate from
 loan_data group by
 OCCUPATION_TYPE order by Default_Rate desc;
14. Find the most common education type among approved applicants.
 select NAME_EDUCATION_TYPE , sum(case when target = 0 then 1 else 0 end) as
 Approve_Applicant from loan_data group by NAME_EDUCATION_TYPE;

15. Determine the average age of applicants who were refused vs. approved.
 select target , round(avg(abs(DAYS_BIRTH)/365.25),1) as Age from loan_data group by target; # use abs for negative no in positive
16. Calculate the default rate for each loan type.
 select NAME_CONTRACT_TYPE , round(sum(case when target=1 then 1 else 0 end)*100/count(*),2) as default_rate
 from loan_data group by NAME_CONTRACT_TYPE;
17. Find the top 5 regions/cities with the highest default rate.
 select REGION_RATING_CLIENT , round(sum(case when target=1 then 1 else 0 end)*100/count(*),2) as default_rate
 from loan_data group by REGION_RATING_CLIENT order by REGION_RATING_CLIENT limit 5;
18. Identify the correlation between credit history length and loan default (e.g., clients with shorter credit histories default more?)
 SELECT
 TARGET,
 ROUND(AVG(ABS(DAYS_EMPLOYED) / 365.25), 1) AS Avg_Credit_History_Years
 FROM loan_data
 GROUP BY TARGET;
19. Determine whether applicants with higher income-to-loan ratio are less likely to default.
 select target , round(avg(AMT_INCOME_TOTAL/AMT_CREDIT),2) as Loan_ratio from loan_data group by target;
20. Find the average number of previous loans for defaulters vs. non-defaulters.
 select target , round(avg(AMT_REQ_CREDIT_BUREAU_YEAR),2) as Avg_Number from loan_data group by target;

 # # AMT_REQ_CREDIT_BUREAU_YEAR consider as a Previous Loan
21. Identify the top 10 high-risk customers based on multiple risk indicators (e.g., low income, high loan amount, poor credit score).
 select AMT_INCOME_TOTAL , AMT_CREDIT ,
 round((coalesce(EXT_SOURCE_1,0) + coalesce(EXT_SOURCE_2,0) + coalesce(EXT_SOURCE_3,0)) / 3 ,3) as Poor_credit from loan_data
 order by AMT_INCOME_TOTAL asc , AMT_CREDIT desc , Poor_credit asc limit 10 ;
22. Create a loan performance report showing for each loan type: total loans, total approved, total defaulted, and average loan amount.
 select NAME_CONTRACT_TYPE , count(*) as Total_Loan , avg(AMT_CREDIT) as Avg_Loan ,
 sum(case when target=0 then 1 else 0 end) as Total_approved , sum(case when target=1 then 1 else 0 end) as total_default
 from loan_data group by NAME_CONTRACT_TYPE;

23. Find the total financial loss due to defaulted loans (based on outstanding amount).
SELECT NAME_CONTRACT_TYPE, SUM(AMT_CREDIT) AS Total_Financial_Loss
FROM loan_data WHERE TARGET = 1 GROUP BY NAME_CONTRACT_TYPE;
24. Calculate the default probability by income group and loan type (using count and ratio).
select NAME_CONTRACT_TYPE ,
case when AMT_INCOME_TOTAL < 50000 then 'Low Income'
WHEN AMT_INCOME_TOTAL between 50000 AND 150000 THEN 'MEDIUM INCOME'
ELSE 'HIGH INCOME' END AS INCOME_GROUP ,
ROUND(SUM(CASE WHEN TARGET = 1 THEN 1 ELSE 0 END),2) AS DEFAULT_LOANS ,
ROUND(SUM(CASE WHEN TARGET = 1 THEN 1 ELSE 0 END) * 100 / COUNT(*), 2) AS
DEFAULT_PROBABILITY FROM LOAN_DATA GROUP BY NAME_CONTRACT_TYPE ,
INCOME_GROUP ORDER BY INCOME_GROUP ASC ;
25. Which gender or marital status group shows a higher approval ratio?
SELECT NAME_FAMILY_STATUS ,
ROUND(SUM(CASE WHEN TARGET = 0 THEN 1 ELSE 0 END) * 100 / COUNT(*),2) AS
APPROVAL_RATIO
FROM LOAN_DATA GROUP BY NAME_FAMILY_STATUS;